Non-parametric spectral estimation techniques: Performance comparison

Performance criteria:

Variability: $\upsilon \stackrel{\text{def}}{=} \frac{V[\hat{P}_x(f)]}{\left(E[\hat{P}_x(f)]\right)^2}$ (small as possible)

Frequency resolution: $\Delta f_{min} \stackrel{\text{def}}{=} (f_2 - f_1)_{min}$ (small as possible)

Ad hoc combined into Figure of merit: $\mathcal{M} \stackrel{\text{def}}{=} \upsilon \cdot \Delta f_{min}$ (small as possible)

	υ (asymptotically)	Δf_{min}	$\mathcal{M} = \upsilon \cdot \Delta f_{min}$
Periodogram	1	$0.89\frac{f_s}{N}$	$0.89\frac{f_s}{N}$
Bartlett	$\frac{1}{K} = \frac{L}{N}$	$0.89\frac{f_s}{L}$	$0.89\frac{f_s}{N}$
Welch (Bartlett window & 50% overlap)	$\frac{9}{8}\frac{1}{K} \cong \frac{9}{16}\frac{L}{N}$	$1.28\frac{f_s}{L}$	$0.72\frac{f_s}{N}$
Blackman-Tukey (Bartlett window)	$\frac{2}{3}\frac{M}{N}$	$1.28\frac{f_s}{2M}$	$0.43\frac{f_s}{N}$

Computational complexity:

Since all methods rely on the FFT, they all have $O(N \log N)$ computational complexity (with different proportionality factors, though)